Operating system and system programming (25CS2104E)

Linux interprocess communication using pipes and FIFOs

Review-I

Under the Esteemed Guidance of

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Abstract

The project **“Linux Inter-Process Communication Using Pipes and FIFOs”** demonstrates how different processes communicate and exchange information in a Linux operating system.
The project mainly focuses on two IPC mechanisms: **Pipes and FIFOs (Named Pipes)**. Pipes are used for communication between related processes, such as a parent and child process, while FIFOs allow communication between independent processes.
Linux system calls such as **pipe(), mkfifo(), read(), write(), and close()** are used to create communication channels and transfer data between processes. The project also demonstrates **process synchronization, data sharing, and file descriptors**.
Overall, this project provides practical knowledge of how Linux manages communication between processes and how Pipes and FIFOs can be used to develop efficient and reliable multi-process applications.

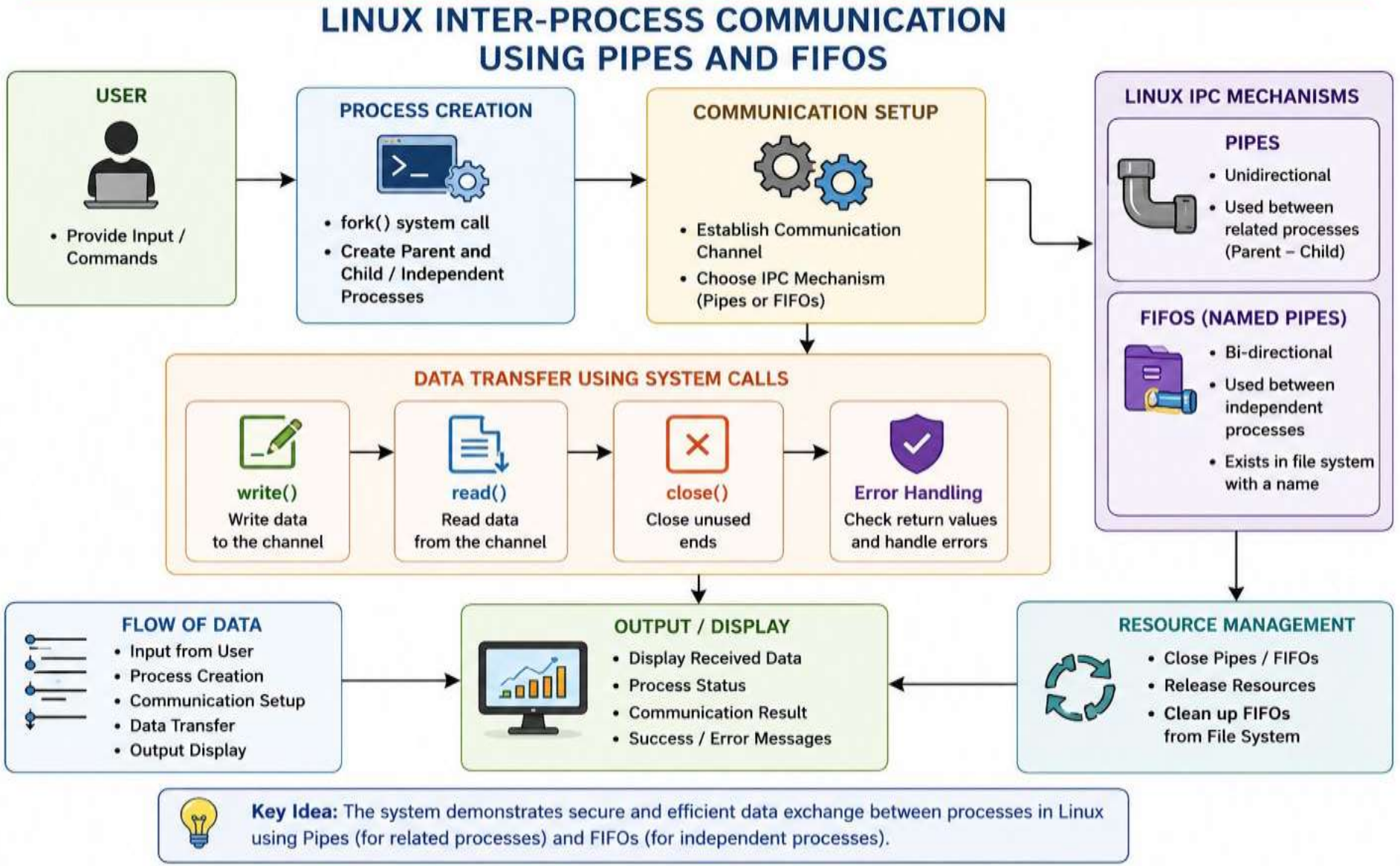
Introduction

- **Inter-Process Communication (IPC)** allows processes in Linux to communicate and exchange data.
- Processes normally have **separate memory spaces**, so an IPC mechanism is needed for data sharing.
- **Pipes** provide communication between related processes, such as parent and child processes.
- **FIFOs (Named Pipes)** allow communication between independent processes.
- This project demonstrates IPC using Linux system calls such as **pipe()**, **mkfifo()**, **read()**, **write()**, and **close()**.

Hardware and Software requirements

Hardware	Software
Processor: Intel Core i3	Operating System: Linux/Ubuntu
Ram: Minimum 4GB	Programming Language: c/c++
Storage: Minimum 20GB free space	Compiler: GCC
Keyboard, Mouse, Display monitor	System Calls: pipe(), mkfifo(),read(),write(),close()

System Architecture



Thank you